

Sustainable development

- 'Sustainability' — keeping an effort going continuously or the ability to last out and keep from falling.
- "the successful management of resources for the development to satisfy the changing human needs while maintaining or enhancing the ability of envt and conserving natural resources."

Strategies for sustainable development

- ① To use locally adaptable, ecofriendly and resource efficient technology, which will use less of resources and produce minimum waste.
- ② To adopt 3-R approach (Reduce, Reuse, Recycle approach) which emphasizes minimization of resource use, using them again and again and recycling the materials so as to decrease pressure on our existing natural resources and reduce generation of waste.
- ③ To promote envtl education and awareness at all levels of education right from school stage to the university level.
- ④ To utilize resources as per carrying of the envt. i.e., consumption of resources should not exceed their regeneration and changes should not be allowed to occur beyond the tolerance capacity of the envt.

Carrying capacity has two basic components.

- ① Supporting Capacity — Capacity to regenerate.
- ② Assimilative Capacity — Capacity to tolerate diff. stress.

→ The issue of sustainable development was discussed in detail for the first time at UN conference on Env't and Development held at Rio-de-Janeiro in 1992. — The Earth Summit.

Its declaration aims at "a new and equitable global partnership through the creation of new levels of cooperation among states".

→ Agenda 21 is one of its five significant agreements, which proposes a global programme of action on sustainable development for the 21st Century in social, economic and political spheres.

2 Aspects of S.D.

① Inter-generation equity.
i.e. to hand over a safe, healthy and resourceful env't to our future generation.

② Intra-generational equity.
i.e. technological developments which promote economic growth of the poor countries so as to narrow wealth gap b/w diff. nations.

Envrtal Ethics and Resource use

The envrtal ethics or Dharma of Ecology teaches us how to treat nature. It includes cultures and traditions that have influenced human relationship with nature, the place of man in nature, rights and duties of individuals, the moral standings of humans and duties of future generations. The basic ~~components~~ principles are.

- (1) protection of renewable natural resources.
- (2) Conservation of non-renewable resources.
- (3) Optimum resource use by nations.
- (4) Avoid consumerism.
- (5) Adopt sustainability as a way of life.
- (6) Joining envt and economics to eradicate poverty.
- (7) Link envt with peace and security.

Environment Laws and Protection Act.

- India is the first country in the world to have provisions for the protection and conservation of envt in its constitution.
- The provision for envtl protection in the constitution were made after the 'UN' conference on Human Environment' held in Stockholm on 5th June, 1972. There after June 5th is celebrated as World Env't Day all over the world. The provision were made through 42nd Amendment of the constitution in 1976. They are.
 - (i) Article 48-A — It declares that "the state shall endeavour to protect and improve the envt and to safeguard forests and wild life of the country".
 - (ii) Art 51 A(g) — declares that "It shall be the duty of every citizen of India to protect and improve the natural envt including forests, lakes, rivers and wildlife and to have compassion for living creatures".
Before 1976, envt was in the state list but after 42nd amendment, it was included in concurrent list.
 - (iii) Art 47, 48 and 51A — According to these articles, It is the duty of state to protect and improve the envt and duty of state to protect and improve the envt and public health and provide pollution free water, air and envt for the public.

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The Environment (Protection) Act 1986.

→ This act came into force on Nov 19, 1986, the birth Anniversary of our late P.M Indira Gandhi. She took keen interest in envtl issues of the country. This act aimed to protect and improve the quality of envt by Central govt with coordination of the State govt.

Powers and Function of Central Govt.

- (i) To set up the Stds of quality of air, water or soil for various and purposes.
- (ii) To set up maximum permissible limits of Conc. of various envtl pollutants (including noise) for diff. areas.
- (iii) To lay procedures and safeguards for the handling of hazardous substances.
- (iv) To set up prohibition and restrictions of the handling of hazardous substances in diff. areas.
- (v) To restrict the areas where industry, operations or processing shall not be allowed.
- (vi) To examine manufacturing processes, materials and substances with which are likely to cause envtl pollⁿ.
- (vii) Establishment and recognition of envtl laboratories.

Functions of State government.

→ Under the Environmental (protection) Rules, 1986 the State pollution Boards have to follow the guidelines provided Under Schedule VI.

Guidelines are

- (i) Advise industries in the State to treat their wastes before their release in the envt.
- (ii) Encouraging industries to recycle and reuse their waste.
- (iii) Promoting implementation of clean technologies by the industries in order to increase fuel efficiency and reduce the generation of envt pollution.
- (iv) Granting permission of the discharge of effluents and emissions into the envt only after assessing the assimilative capacity of the receiving atmosphere/water body.

→ Env Protection Act 1986 was amended in 1994 to include Env Impact Assessment (EIA) of various developmental projects.

The water (Prevention & Control of Pollution) Act 1974.
(Amended 1988)

The water Act was enacted under article 252 (1) of the constitution as a social welfare measure.

- (i) To Prevent and control water pollⁿ.
- (ii) To establish Central and State boards for the prevention and control of water pollution.
- (iii) To provide and confer powers and functions to the Central and State boards to restore wholesomeness of water.

Functions of Central Board.

- To advise the Central Govt. on matters concerning with prevention and control of water pollution.
- To provide technical assistance and guidance to State Boards and sponsor investigations and research relating to water pollution.
- To plan and organize the training of persons engaged in prevention, control or abatement of water pollution.
- To collect, compile and publish technical and statistical data relating to water pollution.
- To lay down the stds for a stream or well waters.
- To ~~plan~~^{plan} and organize nation - wide programme for the prevention and control of water pollution.
- To ~~plan~~ establish or recognise laboratories for analysis of water samples / trade effluent / any sewage.

Functions of State Board.

- To advise the State Govt on matters relating to prevention and abatement of water pollution.
- To plan a comprehensive programme for the prevention, and control or abatement of water pollution.
- To encourage and conduct investigations and research relating to problems of water pollution.
- To collaborate with the Central board in organizing training programmes relating to the control of water pollution and to organize related mass education programme.

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- To lay down effluent standards to be complied with by persons while discharging sewage or sludge or both.
- To advise the State Govt. with respect to the location of any industry which is likely to pollute a stream or well.
- To establish or recognize laboratories for analysis of water of any sewage or trade effluents.

The Air (Prevention & Control of Pollution) Act 1981.
(Amended 1987)

This act was enacted by the parliament to implement the decisions taken at the UN Conference on the Human Environment held at Stockholm in June, 1972.

Objectives are.

- (i) To prevent and control air pollution.
- (ii) To establish Central and State Boards for the prevention and control of air pollution.
- (iii) To provide and confer powers and functions to the Central and State Boards to restore wholesomeness of air.

Functions of Central Boards.

- To advise Central Govt and State boards about improvement in the quality of air and issues related to air pollution.
- To plan and supervise implementation of nation wide programme for prevention and abatement of air pollution.

- To provide technical assistance and guidance to State Boards and industries for prevention and control of air pollution.
- To utilize the services of mass media in educating public about the causes, effects, prevention and control of air pollution.
- To lay down stds for air quality.
- To establish or earmark laboratories for the analysis of air samples.
- To plan and organize training of persons in the field of air pollution.
- To collect, compile and publish technical and statistical data relating to air pollution.

Functions of State Board

- (i) To advise State Govt. on matters concerning air pollution.
- (ii) To plan programmes for prevention and control or abatement of air pollution.
- (iii) To identify the areas or industries causing or likely to cause air pollution.
- (iv) To inspect air quality in air quality in consultation with central board.
- (v) To collect and disseminate information related to causes, prevention and control of air pollution.

Wildlife (Protection) Act 1972 (Amended 1991)

This act safeguards wild animals, birds and plants. Under the Wildlife Protection Act, Indian board of Wildlife was established which is chaired by the Prime Minister of India, and Wildlife advisory boards were constituted for each state.

Objectives.

- Restriction and prohibition on hunting and trapping of wild life.
- Protection of rare and specified plant species.
- Preparation of lists of endangered, rare and threatened species by the Botanical survey of India (BSI) and Zoological Survey of India (ZSI)
- Preservation of biological diversity by setting up and managing National Parks, Wildlife Sanctuaries and Biosphere reserves.
- Control of trade and commerce in some wildlife sps; wildlife products and trophies.
- Setting up of guideline for issuing license for arms in surrounding areas of wildlife sanctuaries.

Forest Conservation Act 1980 (Amended 1988).

Forest Conservation Act was enacted to safeguard forests and their conservation. The important provisions are -

- (1) To check deforestation which causes ecological imbalance and leads to environmental deterioration.
- (2) Put restrictions on the use of forest land for non forest purpose or conversion of reserved forest into non-reserve forest, without a prior approval of the Central govt.
- (3) Provides provision of compensatory afforestation of an equivalent area of non-forest land or double the area of degraded forest for the diversion of a forest land.
- (4) Exercises control over shifting cultivation and encroachments on the forest lands.
- (5) Provides protections to hilly catchment areas, steep slopes and other areas which are under erosion and land slides, and forest areas to save natural biodiversity.
- (6) Puts restrictions on felling of trees above 1000m altitude.
- (7) Provides provision for the constitution of an advisory committee to advise the Central govt with regard to matters connected with conservation of forests.
- (8) Empowers the State govt to declare any forest or wasteland as reserve forest where grazing and felling is not allowed.

Human Population Growth.

Population Explosion → The dramatic increase in population size over a relatively short period is called population explosion.

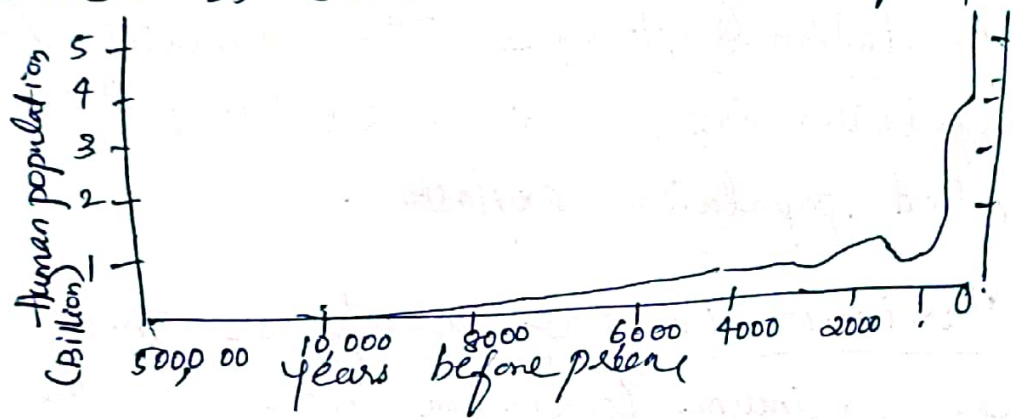
Maximum Carrying Capacity — There is a limit to the maximum population size (ie, the no. of individuals) that can be supported with a given space and resource base. The maximum population size that can be supported by the environment is called the maximum carrying capacity. The environment has the following 3 components:

- (1) Productive system → such as croplands, orchards etc. and provides food and fibre.
- (2) Protective system → such as climate forests, oceans, etc. It buffers air and water cycles, moderates extremes in temp etc.
- (3) Waste assimilative systems → water ways, wetlands etc. that assimilate the wastes produced by human activities.

→ The maximum carrying capacity of the envt depends on the productive and protective life supportive capacity (Productive system and protective system) and waste assimilative capacity. Therefore, the population size should not exceed the maximum carrying capacity and the utilisation of resources should be that lasting damage to the

envt does not occur.

→ Exponential growth → "the increase in a ^{population} nearly is a fixed proportion of its own size during any period of time. It shows a 'J' shaped growth curve.



Population growth rate

It is measured as the annual average growth rate.

$$\text{Average annual growth rate (in \%)} = \left(\frac{P_2 - P_1}{P_1 \times N} \right) \times 100$$

Where P_1 → Population size in the previous census.

P_2 → " " Present census.

N → No. of years b/w the two census.

→ The time required for a population to double itself is called doubling time.

→ Annual average growth rate and doubling time are the two imp indicators of the pace of population growth.

→ Growth rate depends on several factors such as rates of birth, death and migration, age-sex ratio.